Numerical Reproduction and Validation of Forced Convective Cooling of a Heated Obstacle in a Channel: Case 5 Geometry at $k_s/k_f = 10$

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Abstract

This work presents a focused numerical reproduction and validation of a specific benchmark configuration from the study of Young and Vafai [*Int. J. Heat Mass Transfer*, 41 (1998) 3131–3148] on forced convective cooling of a heated obstacle mounted on a channel wall. The reproduction targets Case 5 of the original study ($w/H = h/H = 0.25$) with a solid-to-fluid thermal conductivity ratio of $k_s/k_f = 10$, across the full Reynolds number range investigated in the reference work, $200 \leq Re_{D_h} \leq 2000$. A two-dimensional conjugate heat transfer model was developed in ANSYS Fluent, employing a laminar, steady, incompressible flow solver coupled with a solid-fluid energy equation across the obstacle interface. A user-defined parabolic inlet velocity profile was implemented to match the reference boundary conditions, and results were post-processed and non-dimensionalized to allow direct comparison with the reference correlation and reported trends. The computed mean Nusselt numbers show excellent agreement with the empirical correlation of Young and Vafai (Eq. 16, Case 5 constants), with a mean absolute percentage error of approximately 1.15% across the full Reynolds number range. Local Nusselt number distributions along the obstacle periphery reproduce the characteristic features reported in the original study, including the corner-driven Nusselt number peak at the leading top corner and the secondary peak at the trailing top corner. The peak obstacle temperature is shown to decrease monotonically with increasing Reynolds number, consistent with enhanced convective cooling. This study demonstrates that a scoped, single-geometry, single-conductivity-ratio reproduction of a well-defined benchmark case can achieve high-fidelity quantitative agreement with previously published results.

Keywords: Conjugate heat transfer; forced convection; heated obstacle; channel flow; Nusselt number; CFD validation

1 Introduction

Fluid flow and heat transfer around bluff bodies mounted on channel walls represent a canonical problem in the thermal management of electronic packages, where discrete heat-generating components are cooled by forced convective air flow [1]. Young and Vafai [1] performed an extensive parametric numerical study of this problem using the finite element method, examining the effects of obstacle geometry (six cases spanning combinations of height and width), solid-to-fluid thermal conductivity ratio ($1 \leq k_s/k_f \leq 6600$), Reynolds number ($200 \leq Re_{D_h} \leq 2000$), and heating method (surface flux versus volumetric generation) on the local and mean Nusselt number distributions.

Given the scale of the original parametric study, a complete reproduction of all reported cases is impractical outside of a dedicated, long-term research effort. The present work therefore adopts a scoped, well-defined reproduction strategy: reproducing **Case 5** of the reference study ($w/H = h/H = 0.25$, a square obstacle of intermediate size) at a single, representative thermal conductivity ratio ($k_s/k_f = 10$, corresponding to conductivities typical of encapsulants and epoxy-based materials in electronic packaging), across the *entire* Reynolds number range investigated in the original study. This configuration corresponds directly to Section 3.1 of Young and Vafai [1], in which the effects of Reynolds number variation on the flow field and local Nusselt number distribution are examined in isolation from other parametric variables.

The objective of this study is threefold: (1) to reproduce the governing physics and boundary conditions of the reference problem using an independent CFD solver (ANSYS Fluent) and mesh strategy, (2) to validate the computed mean and local Nusselt number results against the empirical correlation and reported trends of the reference study, and (3) to document a rigorous, transparent methodology for non-dimensionalization and post-processing suitable for direct comparison with a dimensionless reference formulation.

2 Problem Description and Governing Equations

2.1 Geometry

The computational domain consists of a two-dimensional parallel-plate channel of height H containing a rectangular, conducting obstacle mounted on the lower wall (Fig. 1). The obstacle dimensions correspond to Case 5 of the reference study:

$$\frac{w}{H} = 0.25, \qquad \frac{h}{H} = 0.25 \tag{1}$$

An inlet section of length $L_e = 2H$ precedes the obstacle, and an extended outlet section of length $L_o = 8H$ follows it, matching the domain extents specified in the reference study to eliminate entrance and exit effects on the obstacle-region solution.

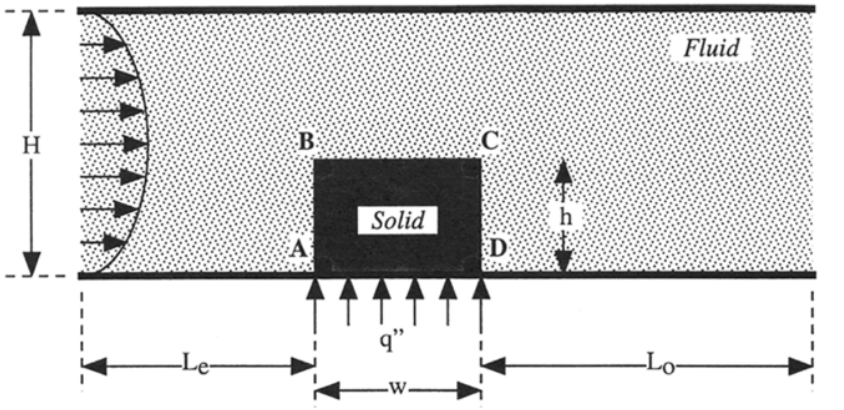


Figure 1: Schematic of the channel and obstacle configuration (Case 5): $w/H = h/H = 0.25$, with inlet length $L_e = 2H$ and outlet length $L_o = 8H$. A parabolic velocity profile is imposed at the inlet, the channel walls are adiabatic, and a constant heat flux q'' is applied at the obstacle base. Points A, B, C, and D denote the leading-bottom, leading-top, trailing-top, and trailing-bottom corners of the obstacle, respectively, and are used as reference locations for the local Nusselt number distribution discussed in Section 4.2.

Table 1: Physical dimensions of the computational domain.

Parameter	Symbol	Value
Channel height	H	0.1 m
Obstacle width	w	0.025 m
Obstacle height	h	0.025 m
Hydraulic diameter	$D_h = 2H$	0.2 m
Inlet length	L_e	0.2 m
Outlet length	L_o	0.8 m

2.2 Governing Equations

The flow is assumed steady, laminar, viscous, incompressible, and Newtonian, with constant thermophysical properties and negligible buoyancy and viscous dissipation. Under these assumptions, the governing equations in dimensionless form [1] are:

Mass conservation:

$$\frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} = 0 \tag{2}$$

Momentum conservation:

$$Re_{D_h}\left(u^*\frac{\partial u^*}{\partial x^*} + v^*\frac{\partial u^*}{\partial y^*}\right) = -\frac{\partial p^*}{\partial x^*} + \left(\frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}}\right) \tag{3}$$

$$Re_{D_h}\left(u^*\frac{\partial v^*}{\partial x^*} + v^*\frac{\partial v^*}{\partial y^*}\right) = -\frac{\partial p^*}{\partial y^*} + \left(\frac{\partial^2 v^*}{\partial x^{*2}} + \frac{\partial^2 v^*}{\partial y^{*2}}\right) \tag{4}$$

Fluid energy conservation:

$$Pe_{D_h}\left(u^*\frac{\partial \Theta_f^*}{\partial x^*} + v^*\frac{\partial \Theta_f^*}{\partial y^*}\right) = \frac{\partial^2 \Theta_f^*}{\partial x^{*2}} + \frac{\partial^2 \Theta_f^*}{\partial y^{*2}} \tag{5}$$

Solid energy conservation (Laplace equation):

$$\frac{\partial^2 \Theta_s^*}{\partial x^{*2}} + \frac{\partial^2 \Theta_s^*}{\partial y^{*2}} = 0 \tag{6}$$

The non-dimensional variables are defined as:

$$x^* = \frac{x}{H}, \quad y^* = \frac{y}{H}, \quad u^* = \frac{u}{U_m}, \quad v^* = \frac{v}{U_m}, \quad p^* = \frac{pH}{\mu_f U_m}, \quad \Theta^* = \frac{T - T_e}{q'' H/k_f} \tag{7}$$

$$Re_{D_h} = \frac{\rho_f U_m D_h}{\mu_f}, \qquad Pe_{D_h} = \frac{\rho_f c_{p,f} U_m D_h}{k_f} \tag{8}$$

2.3 Boundary Conditions

A fully developed parabolic velocity profile is imposed at the channel inlet:

$$u(y) = 6U_m \frac{y(H-y)}{H^2}, \qquad v = 0, \qquad \Theta_f = 0 \tag{9}$$

Zero-gradient conditions are applied at the outlet for velocity and temperature. The channel walls are adiabatic and no-slip everywhere except at the obstacle base, where a constant heat flux $q'' = 10\,\mathrm{W/m^2}$ is applied. At the solid-fluid interface, continuity of temperature and heat flux is enforced:

$$T_f = T_s, \qquad k_f \frac{\partial T_f}{\partial n} = k_s \frac{\partial T_s}{\partial n} \tag{10}$$

2.4 Nusselt Number Definition

The local Nusselt number is defined based on the channel height as the characteristic length:

$$Nu_x = \frac{h_c(x)H}{k_f} = \frac{-1}{\Theta_w}\frac{\partial \Theta_f}{\partial n} \tag{11}$$

Mean Nusselt numbers for each exposed obstacle face (left, top, right) are computed as area-weighted averages:

$$\overline{Nu}_i = \frac{\int_{A_i} Nu_x \, dA}{A_i} \tag{12}$$

and the overall obstacle mean Nusselt number is the area-weighted average of the three exposed faces.

3 Numerical Methodology

3.1 Solver and Discretization

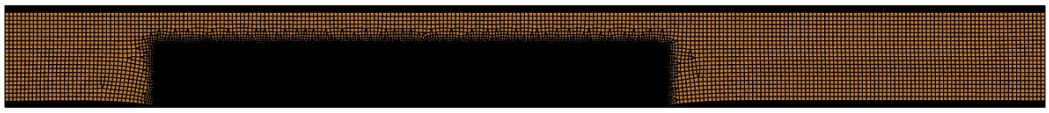
Simulations were performed using ANSYS Fluent with the following configuration:

- **Flow regime:** steady, laminar, incompressible
- **Heat transfer model:** conjugate (solid-fluid coupled energy equation)
- **Pressure-velocity coupling:** Coupled scheme
- **Gradient reconstruction:** Least-Squares Cell-Based
- **Pressure discretization:** Second-order
- **Momentum discretization:** Second-order upwind
- **Energy discretization:** Second-order upwind
- **Convergence criteria:** residuals $< 10^{-6}$ (continuity, momentum), $< 10^{-8}$ (energy)

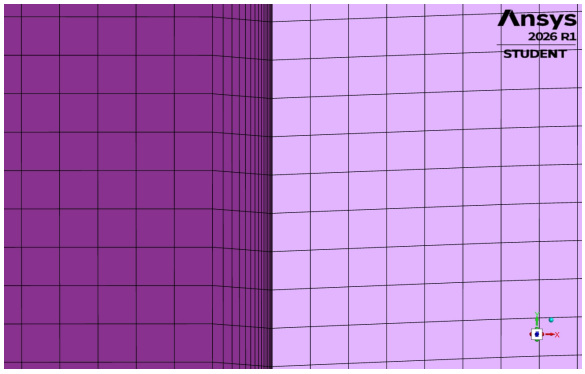
A user-defined function (UDF) was implemented at the inlet boundary to impose the fully developed parabolic velocity profile of Eq. (8), consistent with the reference formulation.

3.2 Computational Mesh

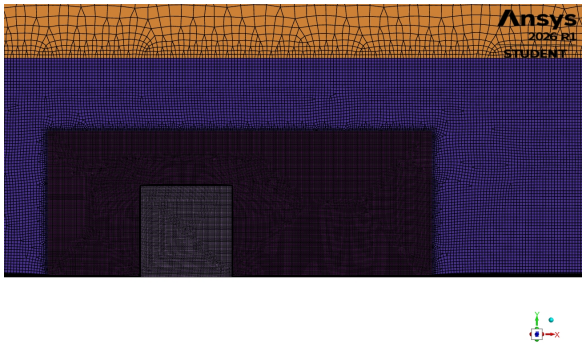
The domain was discretized using a structured quadrilateral mesh comprising 115,933 elements, with 15 layers of boundary-layer elements adjacent to all solid-fluid interfaces (obstacle surfaces and channel walls) to resolve the steep velocity and thermal gradients characteristic of this class of problem. A grid independence test was performed at $Re_{D_h} = 1000$, $k_s/k_f = 10$, comparing coarse ($\sim$50,000 elements), medium (115,933 elements), and fine ($\sim$200,000 elements) meshes. The mean Nusselt number varied by less than 1% between the medium and fine meshes, confirming mesh independence of the production mesh used throughout this study.



(a) Full computational domain: structured quadrilateral mesh with local refinement around the obstacle and near-wall inflation layers.



(b) Close-up of the obstacle region showing the conjugate fluid–solid interface, boundary-layer inflation cells in the fluid zone, and the internal solid mesh.



(c) Mesh of three fluid region and obstacle region close-up view.

Figure 2: Structured mesh used for the conjugate heat transfer simulations (115,933 elements total).

3.3 Fluid and Solid Properties

Table 2: Thermophysical properties and geometric parameters.

Quantity	Symbol	Value
Channel height	H	$0.1\,\mathrm{m}$
Obstacle width	w	$0.025\,\mathrm{m}$
Obstacle height	h	$0.025\,\mathrm{m}$
Inlet length	L_e	$0.2\,\mathrm{m}$
Outlet length	L_o	$0.8\,\mathrm{m}$
Hydraulic diameter	D_h	$0.2\,\mathrm{m}$
Fluid density (air)	ρ	$1.225\,\mathrm{kg/m^3}$
Fluid viscosity (air)	μ	$1.79 \times 10^{-5}\,\mathrm{Pa \cdot s}$
Fluid conductivity (air)	k_f	$0.0242\,\mathrm{W/(m \cdot K)}$
Fluid specific heat	c_p	$973\,\mathrm{J/(kg \cdot K)}$
Prandtl number	Pr	0.72
Solid conductivity ratio	k_s/k_f	10
Solid conductivity	k_s	$0.242\,\mathrm{W/(m \cdot K)}$
Heat flux (obstacle base)	q''	$10\,\mathrm{W/m^2}$
Inlet temperature	T_e	$300\,\mathrm{K}$

The heat flux applied at the obstacle base was $q'' = 10\,\mathrm{W/m^2}$, giving a temperature normalization factor of $q''H/k_f = 41.3223\,\mathrm{K}$. Since the governing energy equation is linear and buoyancy effects are neglected, the dimensionless temperature Θ and Nusselt number Nu are independent of the specific magnitude of the imposed heat flux; the value selected here does not affect the validity of comparison with the reference dimensionless results.

3.4 Reynolds Number and Inlet Velocity

For each target Reynolds number, the mean inlet velocity was computed from

$$U_m = \frac{Re_{D_h}\,\mu_f}{\rho_f D_h} \tag{13}$$

Table 3: Mean inlet velocity for each simulated Reynolds number.

Re_{D_h}	U_m [m/s]
200	0.01461
400	0.02921
600	0.04382
800	0.05843
1000	0.07304
1200	0.08765
1400	0.10225
1600	0.11686
1800	0.13147
2000	0.14608

3.5 Post-Processing

Local Nusselt number distributions were extracted along the obstacle periphery (left, top, and right faces) using Eq. (11), with the peripheral coordinate S^* measuring the running distance clockwise from the lower-left obstacle corner, consistent with the reference study's convention. Mean Nusselt numbers for each exposed face were obtained directly from Fluent's area-weighted surface average reports. All temperatures were non-dimensionalized as

$$\Theta = \frac{T - T_e}{q'' H / k_f} \tag{14}$$

with $T_e = 300\,\mathrm{K}$.

4 Results and Discussion

4.1 Mean Nusselt Number and Correlation Validation

Table 4 summarizes the computed mean Nusselt numbers for the obstacle left ($\overline{Nu}_L$), top ($\overline{Nu}_T$), and right ($\overline{Nu}_R$) faces, the overall obstacle mean ($\overline{Nu}_m$), and the peak obstacle temperature for all ten simulated Reynolds numbers.

Table 4: Computed mean Nusselt numbers and peak temperatures for Case 5, $k_s/k_f = 10$.

Re_{D_h}	U_m [m/s]	$\overline{Nu}_m$	$\overline{Nu}_L$	$\overline{Nu}_T$	$\overline{Nu}_R$	T_{max} [K]	Θ_{max}
200	0.01461	7.1576	8.2841	10.1783	3.0104	302.6178	0.0633
400	0.02921	9.0095	10.3873	13.0032	3.6379	302.2255	0.0539
600	0.04382	10.3038	11.8446	14.8839	4.1830	302.0333	0.0492
800	0.05843	11.2939	12.9949	16.2555	4.6314	301.9162	0.0464
1000	0.07304	12.0825	13.9614	17.2797	5.0062	301.8368	0.0445
1200	0.08765	12.7235	14.8040	18.0395	5.3269	301.7798	0.0432
1400	0.10225	13.2493	15.5558	18.5827	5.6094	301.7371	0.0421
1600	0.11686	13.6822	16.2384	18.9427	5.8655	301.7046	0.0413
1800	0.13147	14.0384	16.8658	19.1443	6.1051	301.6793	0.0413
2000	0.14608	14.3315	17.4482	19.2104	6.3361	301.6594	0.0401

The overall obstacle mean Nusselt number increases monotonically with Reynolds number, approximately doubling from $\overline{Nu}_m = 7.16$ at $Re_{D_h} = 200$ to $\overline{Nu}_m = 14.33$ at $Re_{D_h} = 2000$, consistent with the expected enhancement of convective heat transfer with increasing flow rate. Among the three exposed faces, the top face consistently exhibits the highest mean Nusselt number across the full Reynolds number range, followed by the left face, with the right (downstream) face showing substantially lower values due to its location within the recirculating wake region.

The computed results were compared against the empirical correlation developed by Young and Vafai (Eq. 16 of the reference study) for Case 5:

$$\overline{Nu}_m = a\, Re_{D_h}^{b} \left(\frac{k_s/k_f}{c + k_s/k_f} \right), \qquad a = 1.4537,\ b = 0.3112,\ c = 0.4404 \tag{15}$$

Table 5: Comparison of computed mean Nusselt number with the reference correlation (Eq. 15).

Re_{D_h}	$\overline{Nu}_{m,\mathrm{CFD}}$	$\overline{Nu}_{m,\mathrm{correlation}}$	% Error
200	7.1576	7.2390	−1.12
400	9.0095	8.9820	+0.31
600	10.3038	10.3190	−0.15
800	11.2939	11.2300	+0.57
1000	12.0825	11.9300	+1.28
1200	12.7235	12.5057	+1.74
1400	13.2493	13.0060	+1.87
1600	13.6822	13.4441	+1.77
1800	14.0384	13.8281	+1.52
2000	14.3315	14.1638	+1.18
	Mean absolute % error:		**1.15**

The agreement between the present CFD results and the reference correlation is excellent across the entire investigated Reynolds number range, with a mean absolute percentage error of 1.15% – notably tighter than the $\leq 2.5\%$ mean percent difference reported for the correlation fit itself in the original study. The largest deviations occur at the extremes of the Reynolds number range (200 and 1400–1800), which is consistent with the fact that Eq. (15) is a single power-law fit across the full range and is not expected to capture second-order curvature effects exactly at every point. The systematic, small positive bias for $1000 \leq Re_{D_h} \leq 2000$ suggests a slightly steeper Reynolds number dependence in the present results compared to the fitted exponent $b = 0.3112$, which may be attributable to differences in mesh resolution, discretization scheme (second-order upwind versus the reference finite element formulation), or minor differences in domain truncation.

4.2 Local Nusselt Number Distribution

Figure 3 presents the local Nusselt number distribution along the obstacle periphery (left, top, and right faces) as a function of the non-dimensional peripheral distance S^*, for all ten simulated Reynolds numbers. The peripheral coordinate spans $S^* = 0$ at the lower-left obstacle corner, through $S^* = 0.25$ at the upper-left corner (start of the top face), $S^* = 0.5$ at the upper-right corner (start of the right face), to $S^* = 0.75$ at the lower-right corner.

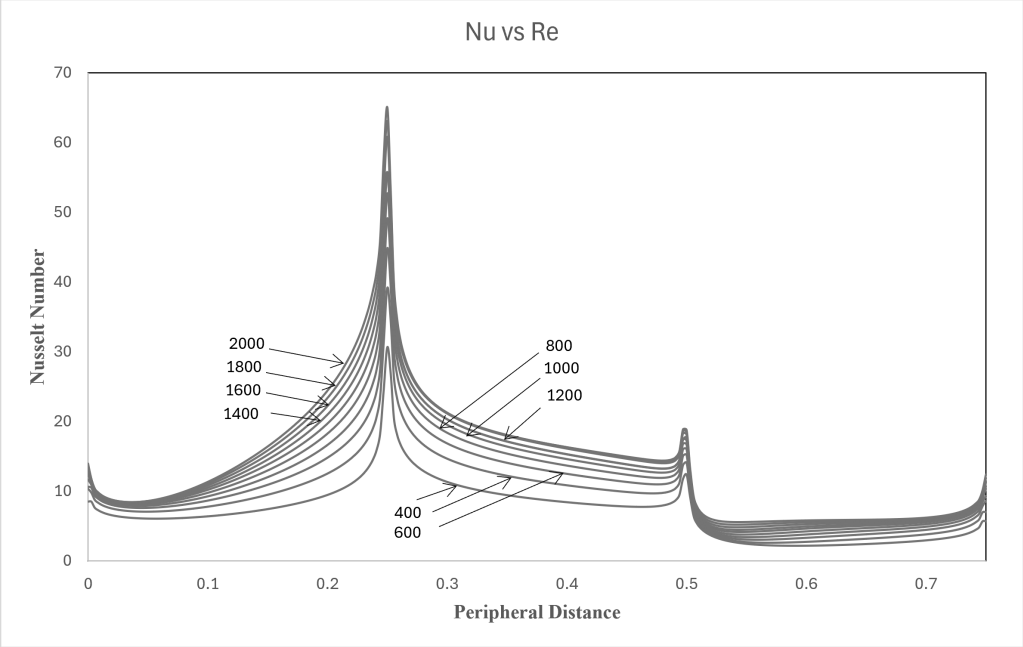


Figure 3: Local Nusselt number distribution around the obstacle periphery for $200 \leq Re_{D_h} \leq 2000$, $w = h = 0.25$, $k_s/k_f = 10$.

The distributions exhibit the characteristic features reported by Young and Vafai [1] for this geometry and conductivity ratio:

1. **Left face ($0 \leq S^* \leq 0.25$):** The local Nusselt number decreases from a moderate value at the lower-left corner to a local minimum near $S^* \approx 0.04$–0.05, reflecting the influence of the weak upstream corner recirculation vortex, before rising sharply as the upper-left corner is approached due to fluid impaction and flow acceleration into the bypass region above the obstacle.

2. **Sharp corner peak at $S^* = 0.25$:** A pronounced, near-singular peak in Nu_x occurs at the upper-left corner, corresponding to the leading edge of the thermal boundary layer developing along the top face. The peak magnitude increases substantially with Reynolds number, from approximately 30.7 at $Re_{D_h} = 200$ to 52.7 at $Re_{D_h} = 1000$, consistent with the thinning of the thermal boundary layer at higher flow rates.

3. **Top face ($0.25 \leq S^* \leq 0.5$):** Following the leading-edge peak, Nu_x decreases monotonically along the top face as the thermal boundary layer develops, characteristic of thermally developing channel flow. A secondary, smaller local rise is observed near the trailing corner ($S^* \approx 0.49$–0.50), attributable to the local reduction in wall temperature as the fluid ceases to be heated beyond the obstacle edge.

4. **Right face ($0.5 \leq S^* \leq 0.75$):** A second, sharper peak occurs immediately at the upper-right corner ($S^* = 0.5$), followed by a rapid decay to small, nearly constant values of Nu_x (in the range of 2–6) along the majority of the downstream face, reflecting the weak, recirculation-dominated heat transfer in the wake region. A modest rise is again observed near the lower-right corner due to the local heat flux boundary condition and proximity to the base.

These qualitative features are in full agreement with the local Nusselt number distributions reported in Fig. 7 of Young and Vafai [1] for the identical geometry, conductivity ratio, and Reynolds number range, providing strong qualitative validation of the present reproduction in addition to the quantitative agreement demonstrated in Table 5.

4.3 Peak Temperature Trends

The maximum obstacle temperature, and its non-dimensional counterpart Θ_{max}, decrease monotonically with increasing Reynolds number (Table 4, Fig. 4), from $\Theta_{max} = 0.0633$ at $Re_{D_h} = 200$ to $\Theta_{max} = 0.0401$ at $Re_{D_h} = 2000$ – a reduction of approximately 36.6%. This trend directly reflects the enhancement in convective heat removal at higher flow rates and is consistent with the fundamental design objective in electronic packaging applications, where increased flow rate is used to reduce peak junction temperatures and improve device reliability.

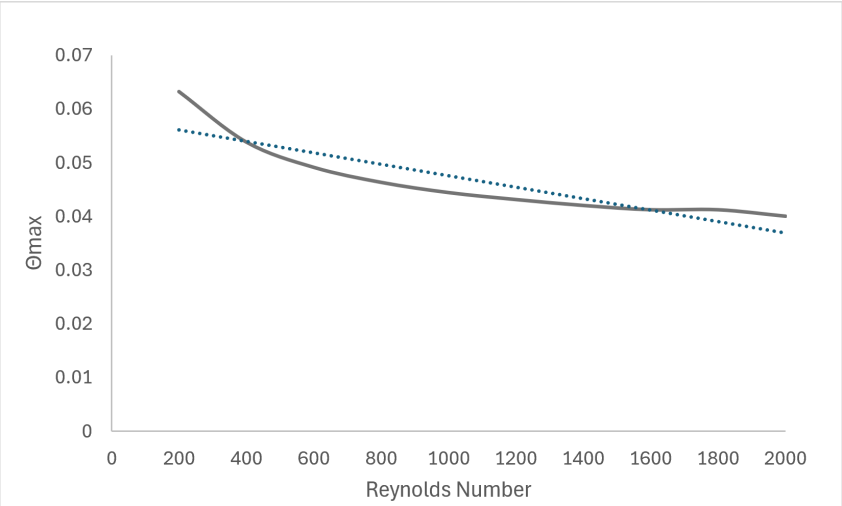
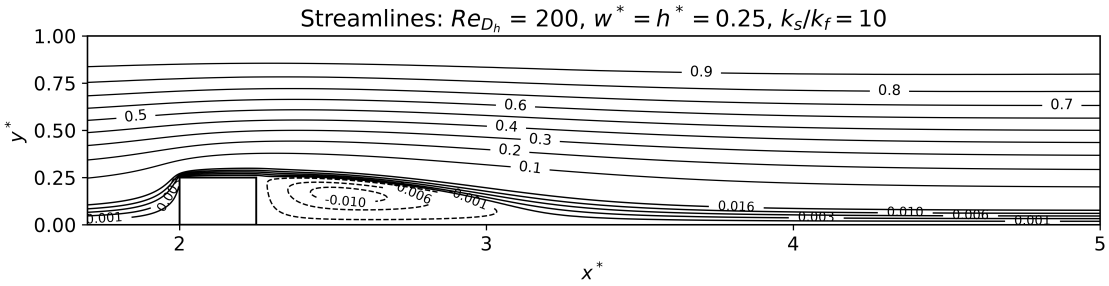


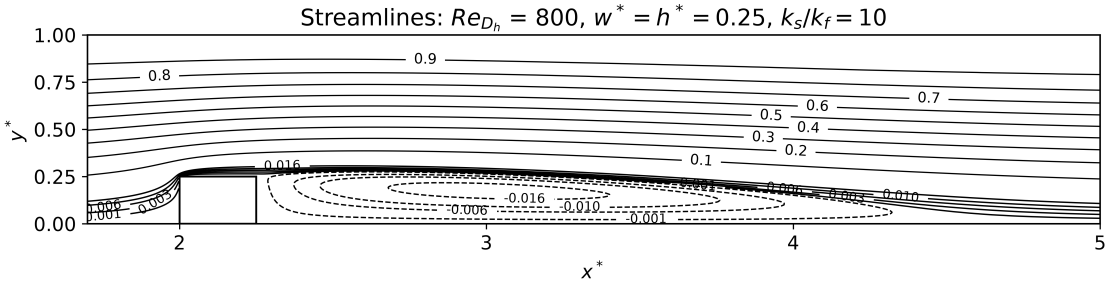
Figure 4: Variation of peak dimensionless obstacle temperature Θ_{max} with Reynolds number, Case 5, $k_s/k_f = 10$.

4.4 Flow Field Characteristics

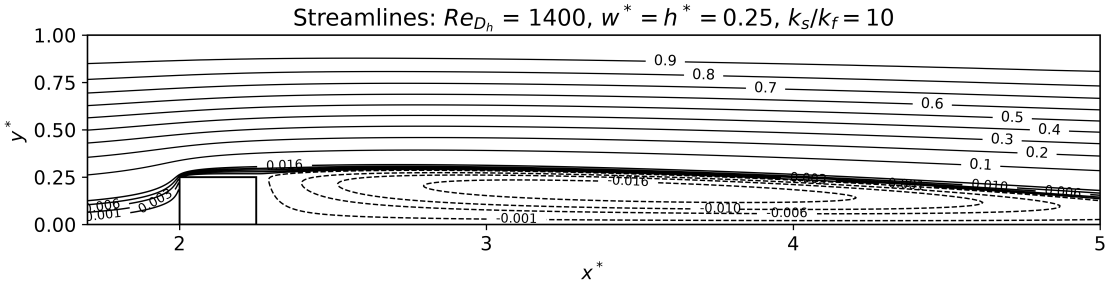
Streamline visualizations at representative Reynolds numbers ($Re_{D_h} = 200, 800, 1400, 2000$) confirm the expected qualitative flow structure reported in the reference study: a weak upstream corner recirculation forward of the obstacle, and a growing downstream recirculation zone whose length increases with Reynolds number due to increased fluid momentum opposing re-expansion into the full channel cross-section (Fig. 5). Isotherm contours for the representative case $Re_{D_h} = 1000$ (Fig. 6) illustrate the conjugate conduction–convection coupling between the obstacle and the surrounding fluid, with a compressed thermal boundary layer at the leading top corner and a downstream thermal wake consistent with the local Nusselt number trends discussed in Section 4.2.



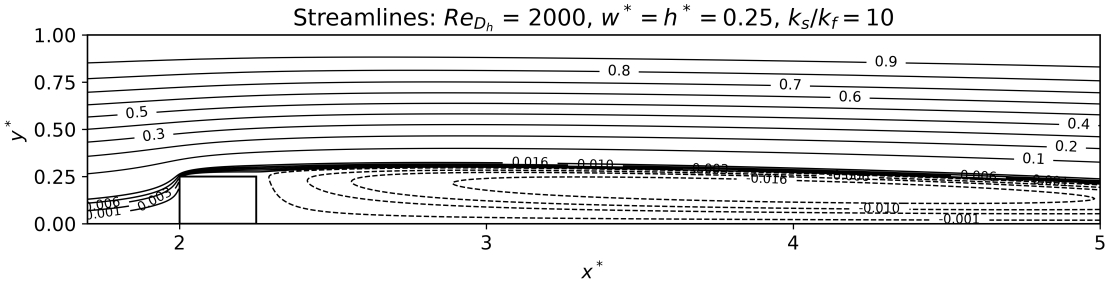
(a) $Re_{D_h} = 200$



(b) $Re_{D_h} = 800$



(c) $Re_{D_h} = 1400$



(d) $Re_{D_h} = 2000$

Figure 5: Stream-function contours ($x^* = x/H$, $y^* = y/H$) showing the growth of the downstream recirculation zone with Reynolds number for Case 5 ($w^* = h^* = 0.25$), $k_s/k_f = 10$. Solid lines: primary flow ($\psi^* > 0$); dashed lines: recirculating flow ($\psi^* < 0$).

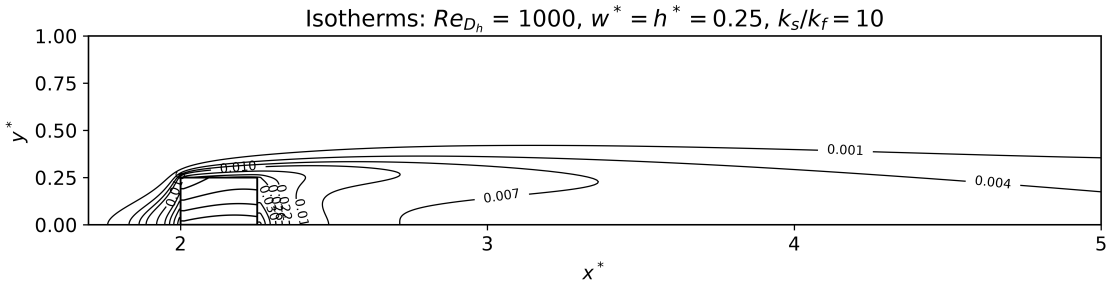


Figure 6: Isotherm (dimensionless temperature Θ) contours in normalized coordinates ($x^* = x/H$, $y^* = y/H$) for Case 5 at $Re_{D_h} = 1000$, $k_s/k_f = 10$. Contours span both the solid obstacle and the fluid domain, illustrating the conjugate conduction–convection coupling at the fluid–solid interface.

5 Summary of Validation

The present reproduction of Case 5 ($w/H = h/H = 0.25$, $k_s/k_f = 10$) from Young and Vafai [1], spanning the complete investigated Reynolds number range ($200 \leq Re_{D_h} \leq 2000$), demonstrates:

- Quantitative agreement with the reference mean Nusselt number correlation to within a mean absolute error of 1.15%, with all individual points agreeing to within 1.9%.
- Qualitative reproduction of all characteristic features of the local Nusselt number distribution along the obstacle periphery, including corner-driven peaks and the thermal boundary layer development along the top face.
- Physically consistent trends of increasing mean Nusselt number and decreasing peak obstacle temperature with increasing Reynolds number.
- Mesh-independent results, confirmed via a dedicated grid convergence study showing less than 1% variation between medium and fine mesh densities.

6 Conclusion

A two-dimensional conjugate heat transfer model reproducing Case 5 of the benchmark study by Young and Vafai [1] was developed and validated using ANSYS Fluent. The scope of this reproduction was deliberately limited to a single obstacle geometry and thermal conductivity ratio ($k_s/k_f = 10$) in order to enable a thorough, high-fidelity investigation across the complete Reynolds number range considered in the original study, corresponding to the parametric analysis presented in Section 3.1 of the reference work. The computed mean and local Nusselt number results show excellent quantitative and qualitative agreement with the reference correlation and reported trends, respectively, validating both the numerical methodology (mesh, discretization, boundary condition implementation) and the post-processing and non-dimensionalization procedures employed. This work illustrates that a well-scoped, rigorously executed reproduction of a single representative case from a larger parametric study can provide meaningful, quantitatively validated results, and represents a suitable foundation for extension to additional geometries, conductivity ratios, or heating methods considered in the original study.

Acknowledgment

The author acknowledges the foundational work of Young and Vafai (1998), whose published methodology and results made this reproduction and validation study possible.

References

[1] T. J. Young, K. Vafai, Convective cooling of a heated obstacle in a channel, *International Journal of Heat and Mass Transfer* 41 (1998) 3131–3148.